Multimodal Immobilization of *Drosophila* Larvae for In Vivo Calcium Imaging

Analysis code for the manuscript: *"Multimodal immobilization of second-instar Drosophila melanogaster larvae using PF-127 hydrogel and diethyl ether for calcium imaging"* (submitted to the Journal of Neuroscience Methods).

About This Code

All analysis code in this repository is **computationally equivalent** to the code used to generate the results reported in the manuscript. The code has been refactored for portability: hardcoded lab paths have been replaced with folder pickers and configuration files, parameters are documented and adjustable, and help documentation has been added. **No computational logic, formulas, thresholds, or statistical methods were changed.** The original lab-internal code is available from the authors upon request.

Repository Structure

```
larval-immobilization-analysis/
│
├── README.md
├── LICENSE
│
├── Motion Analysis/
│   ├── analyze_motion_and_QC/
│   │   ├── analyze_motion_and_QC.m        # Motion tracking for GFP recordings (1 Hz)
│   │   └── motion_analysis_config.m       # Configuration
│   │
│   └── compare_motion_groups/
│       ├── compare_motion_groups.m        # Statistical group comparison of motion
│       └── compare_motion_config.m        # Configuration
│
└── Calcium Imaging/
    ├── analyze_motion_and_QC_neuro/
    │   ├── analyze_motion_and_QC_neuro.m    # Preprocessing: extracts shifts + QC for
pipeline
    │   └── neuro_motion_config.m            # Configuration
    │
    ├── COMPLETE_PIPELINE_RUN_ME_v4_5.m      # ROI detection + trace extraction
    ├── Assess_ROI_Quality.m                 # Signal quality assessment
```

```
├── ROI_REFINEMENT_v2.m              # Multi-criteria ROI quality control
└── COMPARE_NEURO_CONDITIONS_v2.m       # Statistical comparison of conditions
```

Which Scripts Do I Need?

This repository supports two workflows. Use whichever applies to your experiment.

Workflow A: Movement Analysis Only

Use this if you want to **quantify how well an immobilization method works** by tracking residual motion in wide-field fluorescence recordings. This is the workflow that produces the motion comparison results in the paper (Tables 2–4, Figures 2–5).

When to use: You have GFP-expressing larvae recorded at low frame rate (~1 Hz) and want to measure and statistically compare how much they move under different immobilization conditions.

Step 1 — Track motion in each condition's video folder:

```
cfg = motion_analysis_config();
cfg.pixel_size_um = 266/1024;       % your pixel size
cfg.fps = 1;                        % your frame rate
cfg.file_pattern = 'S*.avi';        % your file naming convention
analyze_motion_and_QC('/path/to/control/videos', cfg);
analyze_motion_and_QC('/path/to/experimental/videos', cfg);
```

Step 2 — Compare groups statistically:

```
cfg = compare_motion_config();
cfg.control_label = 'Hydrogel';
cfg.experimental_label = 'Diethyl Ether + Hydrogel';
compare_motion_groups('/path/to/control', '/path/to/experimental', '/path/to/output', cfg);

% To analyze only the first 30 minutes (as in the manuscript's exploratory analysis):
cfg.time_window_min = [0 30];
compare_motion_groups('/path/to/control', '/path/to/experimental', '/path/to/output_30min', cfg);
```

Workflow B: Calcium Imaging Pipeline

Use this if you want to **detect neurons, extract calcium traces, and compare neural activity** between conditions.

When to use: You have GCaMP-expressing larvae recorded at high frame rate (~33 Hz) as BigTIFF files and you want ROIs, calcium traces, and statistical comparisons between conditions.

The calcium imaging workflow has **5 steps.** Each step consumes the output of the previous one:

Step 0: analyze_motion_and_QC_neuro.m → shift CSVs + QC flags (ROI_ELIGIBLE list)
↓
Step 1: COMPLETE_PIPELINE_RUN_ME_v4_5.m → motion-corrected TIFFs, ROIs, ΔF/F traces
↓
Step 2: Assess_ROI_Quality.m → signal quality flags, sample eligibility lists
↓
Step 3: ROI_REFINEMENT_v2.m → refined ROI sets (false positives removed)
↓
Step 4: COMPARE_NEURO_CONDITIONS_v2.m → statistical comparison between conditions

Step 0 — Motion tracking and QC gating (REQUIRED before the pipeline)

This is a **prerequisite** for the calcium pipeline. You must run `analyze_motion_and_QC_neuro.m` on the AVI versions of your neural recordings *before* running the pipeline. This script:

- Performs NoRMCorre rigid registration on the AVI videos to estimate frame-by-frame shifts
- Saves per-frame shift CSVs (`*_shifts.csv`) that the pipeline reads in Step 1 to apply motion correction to the original BigTIFF files
- Produces a dual QC system that determines which files enter the pipeline:
  - `MOVEMENT_VALID` — Is the motion tracking reliable? (registration didn't fail)
  - `ROI_ELIGIBLE` — Is this file suitable for ROI detection? (brain stayed in the field of view)
- Generates an `roi_eligible_list.txt` that the pipeline uses to know which files to process

The pipeline **will not run** without the shift CSVs from this step.

```
cfg = neuro_motion_config();
cfg.pixel_size_um = 266/1024;
cfg.fps_override = 33.33;
```

```
analyze_motion_and_QC_neuro('/path/to/control/avi_videos', cfg);
analyze_motion_and_QC_neuro('/path/to/experimental/avi_videos', cfg);
```

Why AVI and TIFF? The neural recordings exist in two formats: compressed AVI files (smaller, used for motion estimation) and raw BigTIFF files (larger, full quality, used for ROI detection). Step 0 processes the AVIs to compute shifts; Step 1 applies those shifts to the TIFFs.

Step 1 — ROI detection and trace extraction

`COMPLETE_PIPELINE_RUN_ME_v4_5.m` reads the shift CSVs from Step 0 and processes only the files listed in `roi_eligible_list.txt`. For each file it: applies rigid motion correction to the BigTIFF using Bio-Formats, generates a brain mask, computes local correlation (Cn) and peak-to-noise ratio (PNR) maps, detects ROI seeds via adaptive thresholding, grows ROI regions using correlation-based expansion, extracts raw fluorescence traces, applies neuropil subtraction (coefficient = 0.7), computes ΔF/F with a sliding baseline, optionally applies global signal regression to remove shared neuropil signal, smooths traces with a Savitzky-Golay filter, and detects calcium events using GCaMP8f-optimized peak finding.

```
% Edit the config section at the top of the script to set your paths, then:
COMPLETE_PIPELINE_RUN_ME_v4_5();
```

Step 2 — Signal quality assessment

`Assess_ROI_Quality.m` evaluates signal quality across all samples to identify artifacts and neuropil contamination. It categorizes each sample based on pairwise synchrony:

Category	Pairwise Synchrony	Status
`OK`	< 0.15	Clean signal
`MILD_NEUROPIL`	0.15 – 0.30	Acceptable
`HIGH_NEUROPIL`	0.30 – 0.50	Needs regression
`SEVERE_NEUROPIL`	> 0.50	Unusable
`ARTIFACT`	—	Extreme ΔF/F values (> 5 or < −2)

```
% Edit the CONDITIONS and OUTPUT_DIR paths at the top, then:
```

Assess_ROI_Quality;

Step 3 — ROI refinement

`ROI_REFINEMENT_v2.m` applies multi-criteria quality control to remove false-positive ROIs. Criteria include:

- **Shape:** Edge contact (>20% perimeter on brain mask edge), compactness, bounding box fill (<40% or >95%), aspect ratio (>3), solidity (<0.5), eccentricity (>0.95)
- **Trace quality:** Negative transients, rise/decay kinetics inconsistent with GCaMP8f, single-frame step artifacts, sustained plateaus without decay
- **Statistics:** Pairwise trace correlation >0.7 (removes the lower-SNR duplicate), extreme SNR (>150), zero detected events, baseline drift

Each removed ROI is logged with the specific reason for rejection.

% Edit the CONDITIONS paths at the top, then:
ROI_REFINEMENT_v2;

Step 4 — Statistical comparison of conditions

`COMPARE_NEURO_CONDITIONS_v2.m` performs a publication-grade comparison of the *refined* calcium data (post-refinement, not raw pipeline output). Metrics compared include ROI detection quality (yield, removal rate, 11-category removal breakdown), signal quality (SNR, event rates, event amplitudes), motion-induced noise (pairwise synchrony, correlated ROI removal rate), and temporal stability (baseline drift). Statistics include Mann-Whitney U tests, bootstrapped 95% confidence intervals, Cliff's delta and Cohen's d effect sizes, and Bonferroni correction for multiple comparisons.

% Edit the CONDITIONS paths to point to your Refined Data folders, then:
COMPARE_NEURO_CONDITIONS_v2;

Why Are There Two Motion Scripts?

These two scripts share the same core NoRMCorre registration engine but serve completely different purposes:

`analyze_motion_and_QC.m` is the motion **analysis** script. It processes GFP movement recordings (1 Hz, ~60 min) and its outputs — motion metrics like mean speed, median step, and jitter ratio — are the actual data that get statistically compared between conditions using `compare_motion_groups.m`. This is what produces the movement results in the paper.

`analyze_motion_and_QC_neuro.m` is a **preprocessing** step for the calcium pipeline. It processes the AVI versions of neural recordings (33.33 Hz, ~5 min) to extract two things the pipeline needs: (1) per-frame shift CSVs that tell the pipeline how to motion-correct the BigTIFF files, and (2) QC flags (`MOVEMENT_VALID`, `ROI_ELIGIBLE`) that tell the pipeline which files are safe to process. Its motion data is never compared between groups — it exists only to set up the calcium pipeline.

	`analyze_motion_and_QC.m`	`analyze_motion_and_QC_neuro.m`
Role	Analysis endpoint	Preprocessing for calcium pipeline
Input	GFP movement recordings (1 Hz)	GCaMP neural recordings, AVI versions (33.33 Hz)
What happens to its output	Compared between groups	Fed into `COMPLETE_PIPELINE_RUN_ME_v4_5.m`
QC system	Single flag (`QC_PASS`)	Dual flags (`MOVEMENT_VALID` + `ROI_ELIGIBLE`)
Key outputs	Motion metrics for statistical comparison	Shift CSVs + `roi_eligible_list.txt` for pipeline

Installation and Dependencies

MATLAB

MATLAB R2020b or later is required (R2024b recommended). The following toolboxes are needed:

Toolbox	Required by	Notes
Image Processing Toolbox	All scripts	`imtranslate`, `regionprops`, morphological operations
Statistics and Machine Learning Toolbox	Comparison scripts	`ranksum`, `lillietest`, `prctile`
Signal Processing Toolbox	Motion scripts	`medfilt1`, `pwelch` (FFT fallback available if missing)

NoRMCorre

NoRMCorre performs rigid motion registration. Required for both workflows.

git clone https://github.com/flatironinstitute/NoRMCorre.git

Then in MATLAB:

```
addpath(genpath('/path/to/NoRMCorre'));
savepath;  % optional: save so you don't have to add it every session
```

Bio-Formats (Calcium Pipeline Only)

Bio-Formats reads BigTIFF files in the calcium imaging pipeline. The motion analysis scripts do **not** need Bio-Formats.

1. Download the MATLAB toolbox from: https://www.openmicroscopy.org/bio-formats/downloads/
2. Unzip to a permanent location (e.g., `C:\toolboxes\bfmatlab\`)

Add to MATLAB path:

```
addpath(genpath('/path/to/bfmatlab'));savepath;
```

3.

Verify:

```
reader = bfGetReader();   % should not error
```

4.

BigTIFF Writing

The calcium pipeline writes motion-corrected output as BigTIFF files (>4 GB). This uses MATLAB's built-in `Tiff` class with the `'w8'` flag (available since R2014b). No additional installation needed.

Quick Dependency Check

```
% NoRMCorre
assert(exist('normcorre', 'file') == 2, 'NoRMCorre not found. See README for installation.');

% Bio-Formats (only needed for calcium pipeline)
assert(exist('bfGetReader', 'file') == 2, 'Bio-Formats not found. See README for installation.');

% MATLAB Toolboxes
assert(exist('imtranslate', 'file') == 2, 'Image Processing Toolbox not found.');
```

```
assert(exist('prctile', 'file') == 2, 'Statistics and Machine Learning Toolbox not found.');
assert(exist('medfilt1', 'file') == 2, 'Signal Processing Toolbox not found.');

disp('All dependencies OK');
```

Configuration

The motion analysis scripts accept an optional configuration struct. Default parameters match those used in the manuscript. To customize:

```
cfg = motion_analysis_config();    % load defaults
cfg.pixel_size_um = 0.5;           % change what you need
cfg.fps = 2;
analyze_motion_and_QC('/data', cfg);
```

If you omit the config argument, defaults are used and a folder picker dialog opens for path selection.

The calcium pipeline scripts (`COMPLETE_PIPELINE_RUN_ME_v4_5.m`, `Assess_ROI_Quality.m`, `ROI_REFINEMENT_v2.m`, `COMPARE_NEURO_CONDITIONS_v2.m`) use a configuration section at the top of each file. Edit the paths and parameters there before running.

Key Parameters to Set for Your System

Parameter	Where	What it is
`pixel_size_um`	All config files	Pixel size in µm (sensor pixel pitch ÷ magnification)
`fps`	`motion_analysis_config.m`	Frame rate for GFP movement recordings
`fps_override`	`neuro_motion_config.m`	True acquisition rate for neural recordings
`file_pattern`	Config files	Glob pattern matching your video filenames (e.g., `'S*.avi'`)
`config.frameRate`	Pipeline script	Frame rate for calcium imaging (should match `fps_override`)

`config.pixelSize_um`	Pipeline script	Same pixel size as above
`config.conditions`	Pipeline script	Paths to your TIF and AVI data folders

Shift Convention

All scripts use a consistent motion-direction convention:

- `dx_px > 0` = sample moved **right**
- `dy_px > 0` = sample moved **down**

For correction, the pipeline applies `imtranslate(img, [-dx, +dy])`. The sign asymmetry between x and y is intentional: it arises from the conversion between Cartesian coordinates (y-up) used by the motion tracker and image coordinates (y-down) used by `imtranslate`. The column dictionary generated by the pipeline includes a detailed explanation.

Output Files

Motion Analysis Outputs (per folder)

File	Description
`*_shifts.csv`	Per-frame shift estimates (dx, dy in pixels and μm)
`*_motion_trace.png/pdf`	Motion trace visualization
`motion_qc.csv`	QC flags and motion metrics for all files
`motion_enriched.csv`	Extended metrics (PSD, bursts, drift)
`motion_group_summary.csv`	Group-level aggregates
`roi_eligible_list.txt`	Files passing ROI_ELIGIBLE (neuro script only)
`params_motion_*.json`	Full parameter provenance record
`thresholds.tsv`	QC threshold documentation

`column_dictionary_motion.csv`	Column definitions for all output CSVs

Movement Group Comparison Outputs

File	Description
`group_compare_summary.csv`	Statistical comparison (effect sizes, CIs, p-values)
`per_file_metrics.csv`	Per-file metric values for both groups
`*_box_swarm.png/pdf`	Box-swarm plots for each metric
`effects_forest_*.png/pdf`	Forest plot of Hedges' g effect sizes
`conclusions_panel_*.png/pdf`	Full statistical conclusions panel

Calcium Pipeline Outputs

Step 0 — Motion Tracking outputs are listed above (shift CSVs, QC flags, `roi_eligible_list.txt`).

Step 1 — ROI Detection (per file):

File	Description
`*_rigid.tif`	Motion-corrected BigTIFF
`*_rois.mat`	Full ROI data structure (masks, traces, metrics)
`*_roi_metrics_v45.csv`	Per-ROI metrics (SNR, event rate, area, shape)
`*_analysis_v45.png/pdf`	8-panel diagnostic figure
`*_roi_map_v45.png/pdf`	Clean ROI map with numbered outlines

`*_roi_gallery_v45.png/pdf`	Zoomed individual ROI thumbnails
`*_traces_v45.png/pdf`	ΔF/F calcium traces (paginated if >16 ROIs)
`column_dictionary_v45.txt`	Documentation of all CSV columns
`pipeline_summary_v45.csv`	Summary across all processed files

Step 2 — Quality Assessment:

File	Description
`sample_quality_assessment.csv`	All samples with quality categories and metrics
`eligible_samples_*.txt`	Lists of usable samples per condition
`quality_comparison_summary.csv`	Summary statistics across conditions
`quality_diagnostic_figures/`	Per-sample diagnostic plots

Step 3 — ROI Refinement (per file):

File	Description
`*_refined.mat`	Refined ROI data (same structure, fewer ROIs)
`*_refinement_log.csv`	Each removed ROI with the specific rejection reason
`*_rejection_examples.png/pdf`	Visualization showing why ROIs were removed
`refinement_summary.csv`	Cross-condition refinement statistics

Step 4 — Statistical Comparison:

File	Description

`statistical_comparison.csv`	Full statistical results (effect sizes, CIs, p-values)
`forest_plot.png/pdf/svg`	Effect sizes with bootstrap confidence intervals
`removal_breakdown.png/pdf/svg`	ROI removal reasons by condition
`analysis_report.txt`	Comprehensive text report

Reproducing the Manuscript Results

To reproduce the results reported in the paper:

1. **Install dependencies** — MATLAB R2024b, NoRMCorre, Bio-Formats, and the three MATLAB toolboxes listed above.

2. **Set your imaging parameters** — Pixel size (`266/1024` µm for our system) and frame rates (`1` Hz for GFP, `33.33` Hz for calcium).

3. **Movement analysis (Workflow A):**

    1. Run `analyze_motion_and_QC.m` on each condition's GFP video folder
    2. Run `compare_motion_groups.m` for full-duration comparison
    3. Run `compare_motion_groups.m` again with `cfg.time_window_min = [0 30]` for the first-30-minute exploratory analysis
4. **Calcium imaging (Workflow B) — run in order:**

    1. `analyze_motion_and_QC_neuro.m` — produces shift CSVs and `roi_eligible_list.txt`
    2. `COMPLETE_PIPELINE_RUN_ME_v4_5.m` — reads shift CSVs, produces ROIs and traces
    3. `Assess_ROI_Quality.m` — flags artifacts and neuropil contamination
    4. `ROI_REFINEMENT_v2.m` — removes false-positive ROIs
    5. `COMPARE_NEURO_CONDITIONS_v2.m` — statistical comparison on refined data

All default parameters match those used in the manuscript.

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Citation

If you use this code, please cite:

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Contact

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